

Nurture Course

PHASE : SPEED & ACCURACY TEST

TARGET : PRE-MEDICAL 2023

Test Type : **PRACTICE TEST**

Test Pattern : **NEET (UG)**

TEST DATE :

ANSWER KEY

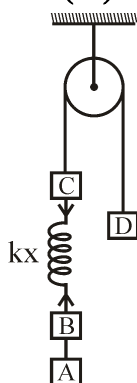
Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	2	3	2	2	3	1	4	1	3	3	3	4	2	4	4	2	3	4	2	2	2	2	3	2	4	1	1	4	2	4
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45															
A.	3	1	2	4	3	1	3	3	4	2	2	1	3	2	2															



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HINT - SHEET

1. **Ans (2)**



spring is initially extended

$$a_A = a_B, a_C = a_D$$

$$kx = (m_A + m_B)g$$

$$\text{so } a_A = a_B = 0 \text{ initially}$$

$$\text{But } a_C = a_D \neq 0$$

$$m_A > m_B > m_C > m_D$$

2. **Ans (3)**

Let mass of sand added be m_1 kg

FBD bucket

$$\Rightarrow 2T = (m + m_1)g$$

$2T$

$(m + m_1)g$

FBD of block

$$\mu Mg \leftarrow \bullet \rightarrow T \Rightarrow T = \mu Mg$$

$$\text{Therefore, } m_1 = 2\mu M - m = 7 \text{ kg}$$

3. **Ans (2)**

For system

$$F_{\text{ext}} = m_1 a_1 + m_2 a_2$$

$$T - (m_1 + m_2)g = m_1 a_1 + m_2 a_2$$

$$T - (10 + 8) \times 10 = 10 \times 2 + 8(0)$$

$$T = 200 \text{ N}$$

4. **Ans (2)**

$$F - \mu N = ma$$

$$\Rightarrow 100 - 0.2(100) = 20a \Rightarrow a = 4 \text{ m/s}^2$$

5. **Ans (3)**

For (1) : m_1 is lifted off if

$$\frac{F}{2} \geq m_1 g \Rightarrow t \geq 20 \text{ s,}$$

For (2) : m_2 is lifted off if

$$\frac{F}{2} \geq m_2 g \Rightarrow t \geq 40 \text{ s}$$

For (3) : At $t = 30$ s acceleration of

$$m_1 = \frac{\frac{F}{2} - m_1 g}{m_1} = \frac{30 - 20}{2} = 5 \text{ m/s}^2$$

For (4) : At $t = 60$ s acceleration of

$$m_2 = \frac{\frac{F}{2} - m_2 g}{m_2} = \frac{60 - 40}{4} = 5 \text{ m/s}^2$$

6. **Ans (1)**

$$\vec{a}_1 = -\vec{a}_2$$

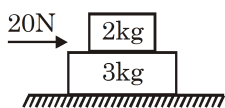
$$\text{So } a_{12} = 2a = 2 \left(\frac{40 - 10}{5} \right) = 12 \text{ m/s}^2 ;$$

$$u_{12} = 0 \text{ so } S_{12} = \frac{1}{2} a_{12} t^2$$

Time to cross to each other

$$6 = \frac{1}{2} \times 12 \times t^2 \Rightarrow t = 1 \text{ sec}$$

7. **Ans (4)**



If both move together



$$a_{\text{Common}} = 4 \text{ m/sec}^2$$

Maximum friction force between 2kg & 3kg

block

$$= \mu mg = 0.3 \times 20 = 16 \text{ N}$$

$$\text{force required to move} = 3 \times 4 = 12 \text{ N}$$

So $F_{\text{requires}} \leq \text{friction force}$.

So both block will move together

8. **Ans (1)**

$$\text{Initially decent of B is } x_1 = \frac{mg}{k_1} = \frac{1g}{10}$$

when mass is placed in pan A.

When, mass is placed in pan B both spring

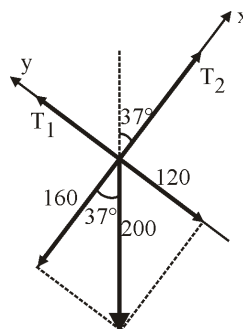
$$\text{extend total extension} = \frac{mg}{k_1} + \frac{mg}{k_2}$$

$$x_2 = \frac{1g}{10} + \frac{1g}{20} = \frac{3}{2} \times \frac{1g}{10} = \frac{3}{2} x_1$$

9. **Ans (3)**

$$\Sigma F_x = 0 \Rightarrow T_2 = 160 \text{ N}$$

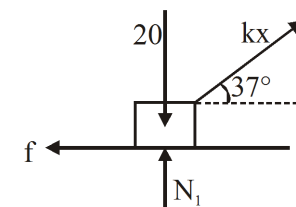
$$\Sigma F_y = 0 \Rightarrow T_1 = 120 \text{ N}$$



Spring force = 120 N

$$\text{Spring constant } k = \frac{120 \text{ N}}{4 \text{ cm}} = 3000 \text{ N/m}$$

10. **Ans (3)**



$$kx = 20 \text{ N}, N_1 = 8 \text{ N},$$

$$f = 4 \text{ N}; a = \frac{16 - 4}{2} = 6 \text{ m/s}^2$$

11. **Ans (3)**

$$100 = k \times (40 - \ell_0), \quad 200 = k \times (60 - \ell_0)$$

$$\Rightarrow 2 = \frac{60 - \ell_0}{40 - \ell_0} \Rightarrow 80 - 2\ell_0 = 60 - \ell_0$$

$$\Rightarrow 20 = \ell_0 \Rightarrow 20k = 100 \Rightarrow k = 5$$

$$\text{Therefore } x = k \times (30 - 20) = 50 \text{ N}$$

12. **Ans (4)**

Net force by surface on block will be opposite gravity.

13. **Ans (2)**

If tension in string connecting m_1 & m_3 is T

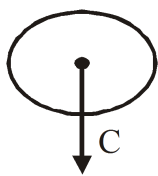
$$\text{Then } T = m_1 g \sin 30^\circ$$

$$2T = m_2 g, T = m_3 g$$

$$\text{so, } \frac{m_1}{2} = \frac{m_2}{2} = m_3$$

$$\text{or } m_1 : m_2 : m_3 :: 2 : 2 : 1$$

14. **Ans (4)**



[contact will be braked when lift move downward]

$$C + mg = ma$$

$$C = 20\text{N} \text{ \& } mg = 1\text{N}$$

$$Q \ C \gg mg \Rightarrow C \approx ma$$

$$a = \frac{C}{m} = \frac{20}{1/10} = 200 \frac{\text{m}}{\text{sec}^2}$$

$$\text{and } a = -200 \frac{\text{m}}{\text{sec}^2} \text{ at } t = 300 \text{ sec}$$

15. **Ans (4)**

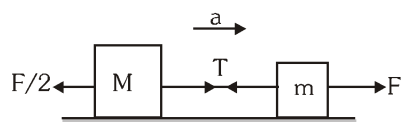
Let acceleration of system be a

then according to question

$$(N-4)ma = 2(N-8)ma \Rightarrow N-4 = 2N-16$$

$$\Rightarrow N = 12$$

16. **Ans (2)**



$$a = \frac{F - F/2}{M + m} = \frac{F}{2(M + m)}$$

$$F - T = ma \Rightarrow T = F - \frac{mF}{2(M + m)} = \frac{(2M + m)}{2(M + m)} F$$

$$\text{If } M = m \text{ then } T = \frac{3F}{4} \text{ \& if } M \gg m$$

then $T \rightarrow F$

17. **Ans (3)**

$$\text{For } 3\text{kg} : a_3 = 0 \Rightarrow \frac{F}{2} = 3g \Rightarrow F = 6g$$

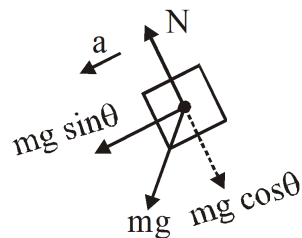
$$\text{For } 2\text{kg} : \frac{\frac{F}{2} - 2g}{2} = a_2 \Rightarrow a_2 = \frac{g}{2}$$

$$\text{Acceleration of pulley, } a_p = \frac{a_2 + a_3}{2} = \frac{g}{4}$$

18. **Ans (4)**

FBD of block

With respect to wedge



$$N = mg \cos \theta$$

$$\text{and } mg \sin \theta = ma$$

$$\text{For (D) : } mg - N \cos \theta = mg - mg \cos 2\theta \\ = mg \sin 2\theta = ma \sin \theta$$

19. **Ans (2)**

Acceleration of system,

$$a = \frac{15t - 3g \sin 53^\circ}{3} = 5t - 8$$

$$\text{For } 2\text{kg block, } T - 2g \sin 53^\circ = 2a$$

$$\Rightarrow 50 - 16 = 2(5t - 8) \Rightarrow t = 5\text{s}$$

20. **Ans (2)**

$$a = \frac{3-2}{3+2}g = 2\text{m/s}^2, \text{ Velocity } v = at$$

$$\Rightarrow 6 = 2t \therefore t = 3$$

21. **Ans (2)**

Let normal force be R

In equilibrium, tension in string is 28N

For system of box 1 & 2

$$\uparrow T = 28\text{N}$$

$$\uparrow R$$

$$\downarrow 90\text{N}$$

$$\Rightarrow R = 62\text{N}$$

22. **Ans (2)**

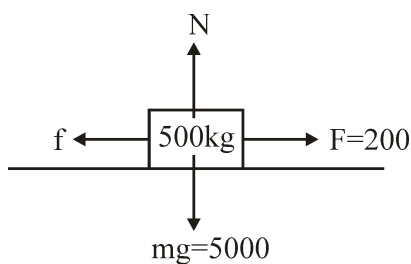
$$\text{Acceleration of box '3' is } 2a = \frac{t}{2}$$

$$\text{For lifting box '2' ; } T \geq 35\text{N}$$

$$\text{For box '3' ; } T - 28 = (2.8) \left(\frac{t}{2} \right)$$

$$\Rightarrow 35 - 28 = 1.4t \Rightarrow t = 5\text{s}$$

23. Ans (3)



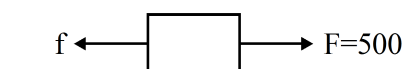
For constant speed

$$f = F$$

$$\mu N = 200$$

$$\mu 5000 = 200$$

$$\mu = \frac{200}{5000} = \frac{1}{25}$$



$$F - f = 500 a$$

$$500 - \mu N = 500a$$

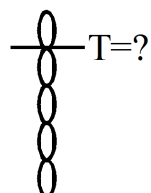
$$500 - \frac{1}{25} 5000 = 500a$$

$$300 = 500a$$

$$a = 0.6 \text{ m/s}^2$$

24. Ans (2)

Given $m = 0.1 \text{ kg}$



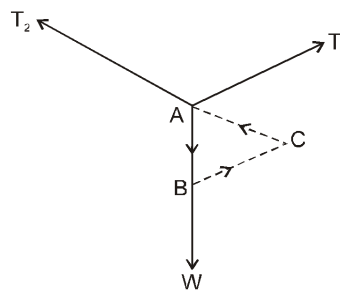
$$a = 1.2 \text{ m/sec}^2$$

$$T = 4 mg + 4ma$$

$$= 4(.1)(g) + 4(.1)(1.2)$$

$$= 4.4 \text{ N}$$

25. Ans (4)



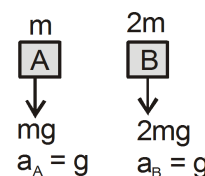
$$\vec{AB} = \vec{W}, \vec{BC} = \vec{T}_1, \vec{CA} = \vec{T}_2$$

$$\vec{AB} + \vec{BC} + \vec{CA} = 0$$

26. Ans (1)

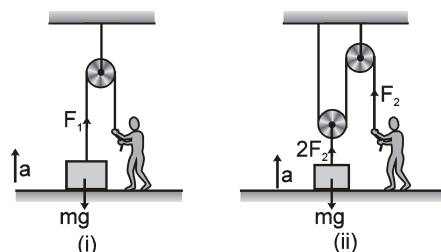
In this case spring force is zero initially

F.B.D. of A and B



27. Ans (1)

Since, $h = \frac{1}{2} at^2 \Rightarrow a$ should be same in both cases, because h and t are same in both cases as given.



$$\text{In (i)} F_1 - mg = ma.$$

$$\Rightarrow F_1 = mg + ma.$$

$$\text{In (ii)} 2F_2 - mg = ma$$

$$\Rightarrow F_2 = \frac{mg + ma}{2} \quad \therefore F_1 > F_2$$

28. Ans (4)

F.B.D. of block

$$2T = 2F = 4mg$$

$$mg$$

$$\Rightarrow 2T - mg = ma$$

$$a = 3g$$

29. **Ans (2)**

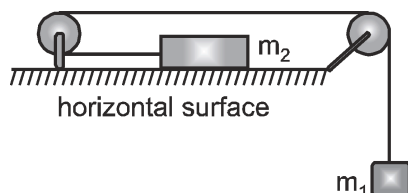
(Acceleration of boy and block will be same

equal to 1.25 m/s^2 w.r.t. ground. Hence

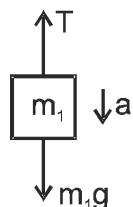
$$10 = \frac{1}{2}(1.25)t^2 \Rightarrow t = 4 \text{ sec.}$$

30. **Ans (4)**

The acceleration of block of mass m_1 is



From FBD of block m_1

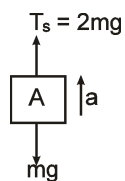


$$m_1g - T = m_1a$$

$$\Rightarrow T = m_1(g - a) = 1 \times 5 = 5 \text{ N}$$

31. **Ans (3)**

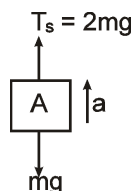
For first case tension in spring will be



$T_s = 2mg$ just after 'A' is released.

$$2mg - mg = ma \Rightarrow a = g$$

In second case $T_s = mg$



$$2mg - mg = 2mb$$

$$b = g/2$$

$$a/b = 2$$

32. **Ans (1)**

$$\text{Let } a = \text{acc}^n \text{ of } m_1 \text{ acc}^n \text{ of pulley} = \frac{a+0}{2} = \frac{a}{2}$$

If acceleration of $m_2 = b$

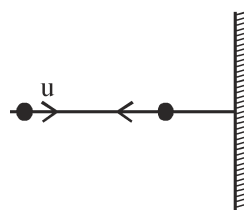
$$\text{Then } 0 + \frac{b}{2} = \frac{a}{2}$$

Hence $a = b$

$$T = m_1 a, m_2g - T = m_2 a$$

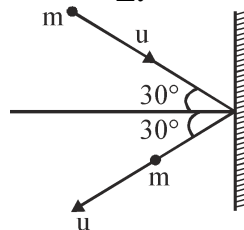
$$\therefore a = \frac{m_2g}{m_1 + m_2}$$

33. **Ans (2)**



$$\Delta \vec{P} = 2mu(-\hat{i})$$

$$\vec{F}_1 = \frac{2mu}{\Delta t}(-\hat{i})$$



$$\Delta P = 2mu \cos 30^\circ(-\hat{i})$$

$$\vec{F}_2 = \frac{mu\sqrt{3}}{\Delta t}(-\hat{i})$$

34. **Ans (4)**

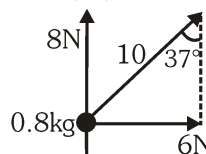
$$a = \frac{4g \sin 30^\circ - 0.5g}{4 + 0.5 + 0.5} = 3 \text{ m/s}^2$$

$$\text{For A} = 4g \sin 30^\circ - T_A = 4 \times 3, T_A = 8 \text{ N}$$

$$\text{For B} = T_B - 0.5g = 0.5 \times 3, T_B = 6.5 \text{ N}$$

$$T_A - T_B = 8 - 6.5 = 1.5 \text{ N}$$

35. **Ans (3)**



$$a = \frac{6 \text{ N}}{0.8} = 7.5 \text{ m/s}^2; \theta = 37^\circ$$

36. Ans (1)

$$a_p = a_{\text{pulley}} = 2 \text{ m/s}^2,$$

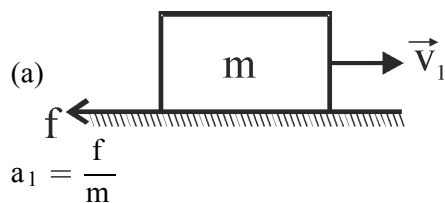
$$a_{cp} = 1 \text{ m/s}^2 \uparrow, \vec{a}_c = \vec{a}_{cp} + \vec{a}_p = 3 \text{ m/s}^2 \uparrow$$

37. Ans (3)

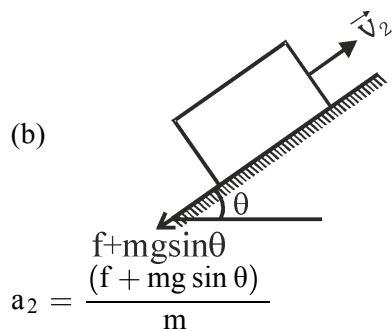
$$T_A - m_1 g - T_B = 2m_1$$

$$6 = m_1 \times 12 \Rightarrow m_1 = 0.5 \text{ kg}$$

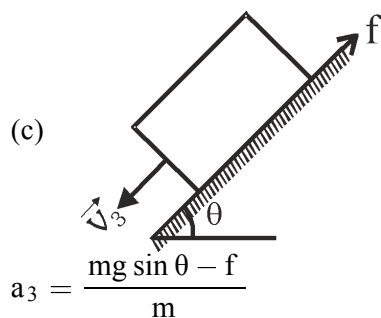
38. Ans (3)



$$0 = v_1^2 - 2a_1 S$$



$$0 = v_2^2 - 2a_2 S$$



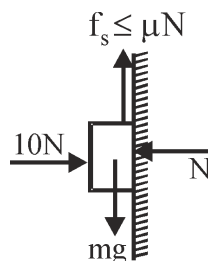
$$0 = v_3^2 - 2a_3 S \quad \text{as} \quad a_3 < a_1 < a_2$$

$$\text{so } v_3 < v_1 < v_2$$

39. Ans (4)

10 N is minimum force to hold stationary,

$$\text{so, } f_s = \mu(10) = 2 \text{ N}$$



$$\text{Contact force } R = \sqrt{N^2 + f^2} = \sqrt{10^2 + 2^2} = 2\sqrt{26} \text{ N}$$

40. Ans (2)

Force is in positive direction where acceleration is positive i.e. slope of v-t graph is positive.

41. Ans (2)

$$T = mg \Rightarrow \text{constant} \quad (m = \text{mass of block})$$

$$f = T \cos \theta$$

As man moves away θ decreases $\Rightarrow f$ increases.

$$N + T \sin \theta = Mg \quad (M = \text{mass of man})$$

$$N = Mg - T \sin \theta$$

as θ decreases N increases.

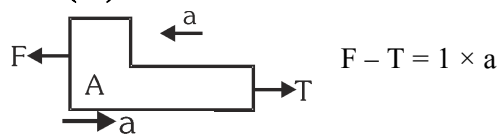
42. Ans (1)

$$F = m \frac{\Delta v}{\Delta t} \Rightarrow 5 = \left(\frac{15}{1000} \right) \times \frac{20 - 0}{t} \Rightarrow t = 0.06 \text{ s}$$

43. Ans (3)

Vertical displacement is same in both cases so both will reach simultaneously if both are released together.

44. **Ans (2)**



(1) $T = 1.5 (1)$

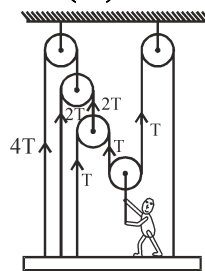
$\Rightarrow a = 1.5 \text{ ms}^{-2}, a_{\text{rel}} = 1.5 + 1.5 = 3 \text{ ms}^{-2}$

here $u_{\text{rel}} = 0$

Hence, $s_{\text{rel}} = 0 + \frac{1}{2} (a_{\text{rel}}) t^2$

$\Rightarrow t = \sqrt{\frac{2s_{\text{rel}}}{a_{\text{rel}}}} = \sqrt{\frac{2 \times 13.5}{3}} = 3 \text{ sec}$

45. **Ans (2)**



FBD of platform + man

$\uparrow 10 T$



$120 g$

$\Rightarrow T = 120 \text{ N}$

$\Rightarrow \text{Force applied by man} = 2T = 240 \text{ N}$